FishONet Challenge

Open-Set Fish Species Recognition — Technical Results Report

CV4Ecology Workshop Competition · Codabench Competition 16815
Single-pipeline compliant system · Measured Codabench evidence only

Best overall	Seen / Unseen	vs visible #1	Gap to 53%
50.756%	78.952% / 14.356%	+0.196 pt	2.244 pt

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Abstract. We present our end-to-end approach and measured results for the FishONet open-set recognition challenge. The evaluation set comprises 35,665 images (20,097 seen / 15,568 unseen) over 17,393 species classes, scored with population-weighted overall accuracy (0.5635·seen + 0.4365·unseen). Our best confirmed Codabench submission (v43 dual BioCLIP-2) reaches **50.7556%** overall (seen 78.952%, unseen 14.356%), exceeding the visible leaderboard #1 overall by +0.196 pt while remaining 2.244 pt short of a 53% target bar. Gains since our early ~45% baseline are driven by a combined image–text novelty gate, routing fraction tuning, Sinkhorn assignment on the unseen route, shift-robust BioCLIP fine-tunes, TaxaBind text fusion, and external iNaturalist prototypes (mean → maxpool → BioCLIP-2 → dual frozen+LoRA). We document submission changelogs with rationales, ablations that failed to transfer, and the remaining unseen headroom.

1. Problem formulation and evaluation protocol

The Fish Species Recognition Challenge requires predicting a species name for every evaluation image among 17,393 candidates. Seen classes have training images; unseen classes are text-only by task definition (no training photographs). Organizers prohibit folder-oracle use of `splits/test.pkl` / `splits/unseen.pkl` to decide membership or shrink the candidate set. Our system therefore applies one uniform rule to all 35,665 images and argmaxes over the full class space.

Official scoring emphasizes overall accuracy. Codabench also reports seen and unseen accuracies. With population weights w_seen=0.5635 and w_unseen=0.4365, overall is dominated by seen accuracy, but competitive movement after ~48% is almost entirely in the unseen term b. Submission quota is 3/day and 30 total — we treat every Codabench slot as expensive evidence.

Quantity	Value	Notes
Eval images	35,665	20,097 seen + 15,568 unseen
Seen classes	5,795	With training images
Unseen classes	11,598	Text-only; no train photos
Total classes	17,393	Full argmax space
Score	Overall accuracy	0.5635·seen + 0.4365·unseen
Quota	3/day · 30 total	Budget via holdout EV + transfer factors
Timeline	1 Jun – 1 Sep 2026	Results 9 Sep 2026

Table 1. Challenge quantities used throughout this report.

2. System overview

Our production stack is a BioCLIP-family vision–language system with shift-robust fine-tuning, concentrated ensembling, taxonomic text prompts (taxctx), database normalization (dbnorm), TaxaBind text features, and iNaturalist-derived class prototypes. A novelty gate routes a fraction f=0.72 of images toward a closed-set-like path and the remainder toward an unseen-oriented path with Sinkhorn-balanced assignment. Compliance is mandatory: one pipeline, full 17,393-way argmax, no fish-specific third-party recognizers, disclosed external text/images.

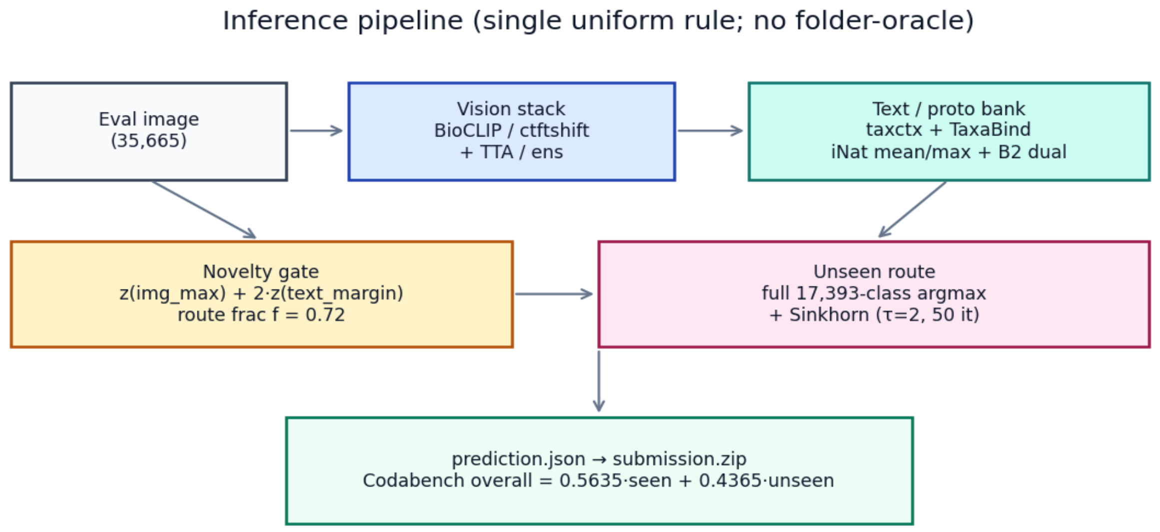

Figure 1. Inference architecture used for all v33+ submissions.

2.1 What consistently worked

- Combined gate z(img_seenmax) + 2·z(text_margin) — gate became net-positive at v23.
- Routing fraction f=0.72 (not the 0.5635 population prior) — v28→v31.
- Sinkhorn on the unseen route (tau=2.0, 50 iterations) — +0.09 pt real at v33; coverage 38.6%→57.5%.
- ctftshift + TaxaBind text — largest single jump (v36: +1.26 overall / +1.91 unseen vs v33).

- iNat mean prototypes (v37), photo maxpool (v40), BioCLIP-2 mean proto (v41), dual frozen+LoRA-v2 BioCLIP-2 (v43).
- dbnorm (dim=0) is load-bearing (–1.60 unseen if dropped).

3. Codabench climb and evidence

Figure 2 plots every major measured overall score from our early gamma baselines (~45.2%) to the current best (v43, 50.76%). Orange markers denote real regressions relative to the contemporaneous best (v34 recover-on-eject; v39 TaxaBind image protos). The red dashed line is visible leaderboard #1 at 50.56%; the green dotted line is our internal 53% target.

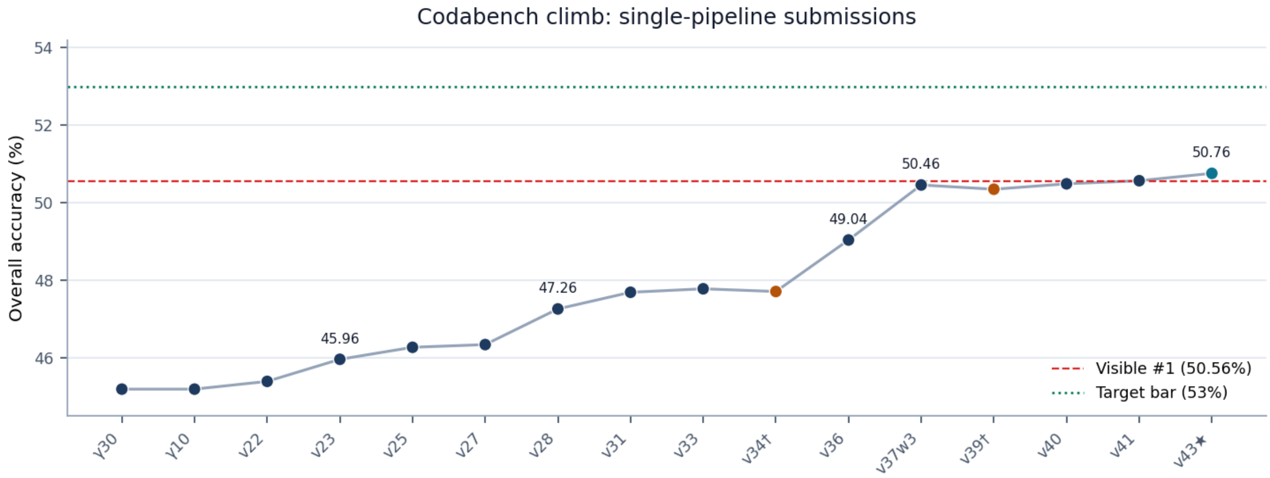

Figure 2. Measured Codabench overall accuracy across submissions.

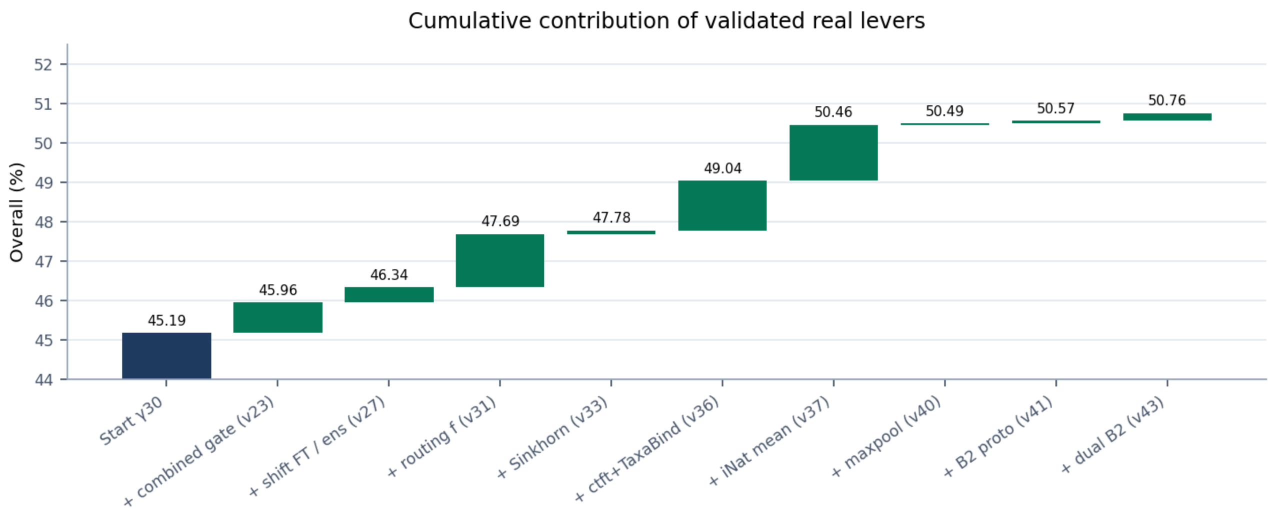

Figure 3. Cumulative contribution of levers that transferred to Codabench (dead levers omitted).

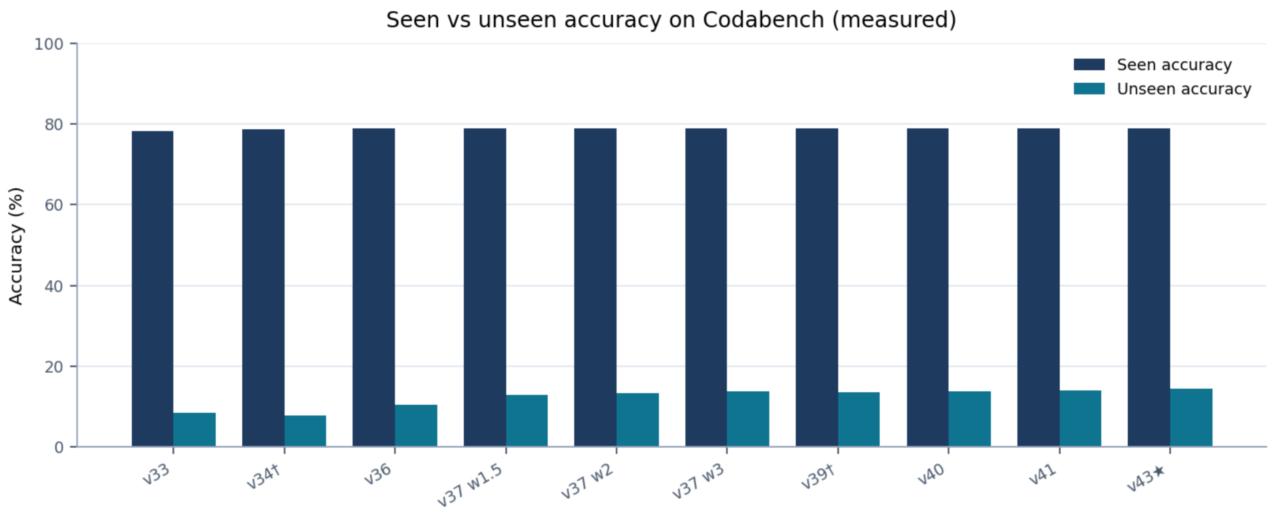

Figure 4. Seen accuracy is essentially flat (~78.95%) after v36; competitive gains are unseen-side.

4. Submission changelog (files, results, and why)

Table 2 lists the principal submission artifacts under `submissions/`, the measured Codabench outcome when submitted, and the scientific rationale. Scores are stated only when measured on Codabench; brackets built but held or projected are marked accordingly.

ID	Artifact	Overall %	Seen/Unseen	Why (change & outcome)
v33 sink72	submission_v33_sink72.zip	47.78	78.20 / 8.51	Add Sinkhorn balanced assignment on unseen route; lift class coverage.
v34 recover	(recover-on-eject)	47.71†	78.63 / 7.80	Holdout +3.51 lied; recovering ejected seen hurt unseen (–0.71). Dead.
v36 ctft+TB	submission_v36_ctftshift_tb_sink72.zip	49.04	78.95 / 10.42	Shift-FT BioCLIP + TaxaBind text; largest single real jump (+1.26).
v37 w1.5	submission_v37_inat_proto_w1.5_sink72.zip	50.12	78.95 / 12.89	iNat mean image prototypes on unseen route; first post-v36 real lift.
v37 w2	submission_v37_inat_proto_w2_sink72.zip	50.32	78.95 / 13.35	Weight sweep; diminishing but positive (+0.20 overall vs w1.5).
v37 w3	submission_v37_inat_proto_w3_sink72.zip	50.46	~13.7 unseen	Higher IMG_W; validated mean-proto anchor before maxpool.
v39 TB-img	submission_v39_inat_tbimg_w3_0.5_sink72.zip	50.35†	78.95 / 13.44	Proxy +0.52 projected gain; real moved opposite (–0.11). Dead.
v40 maxpool	submission_v40_inat_maxpool_w4_sink72.zip	50.49	78.95 / 13.76	Top-k photo maxpool bank; tiny real +0.03; transfer ~0.02.
v41 B2 proto	submission_v41_bioclip2proto_w4_b22.5_sink72.zip	50.57	78.95 / 13.93	BioCLIP-2 iNat mean proto @ B2_W=2.5; +0.08; transfer ~0.08.
v42 LoRA (opt.)	submission_v42_b2lora_v2_w4_b21.5_sink72.zip	proj ~+0.03	—	Optional LoRA-v2 alone; not the 53% path; superseded by dual.
v43 dual ★	submission_v43_b2dual_w4_wf0.5_wl1_sink72.zip	50.756	78.952 / 14.356	Frozen B2 wf=0.5 + LoRA-v2 wl=1.0; best real; transfer ~0.36.
v43 w3 (hold)	submission_v43_b2dual_w3_wf0.5_wl1_sink72.zip	held	—	Proxy weaker than w4 (~44.95 vs ~45.56); do not burn a slot.

Table 2. Submission changelog. † = real regression vs contemporaneous best. ★ = current best.

4.1 Detailed rationale by era

Gate & routing (γ→v31). Early novelty gates were effectively worthless (γ30≈γ10≈45.19%). v23's combined image+text gate made novelty detection net-positive. Raising the seen routing fraction from the population prior to 0.65 then 0.72 (v28/v31) recognized that with weak unseen accuracy, over-ejecting valuable seen images is a bad trade.

Assignment & compliance (v33–v34). Sinkhorn improved unseen coverage with a small but real +0.09 pt. Soft recover-on-eject (v34) is a cautionary tale: holdout gains of +3.51 pt reversed on Codabench (–0.07 overall) because seen recovery came at a larger unseen cost. We thereafter insist on measured transfer factors before spending slots.

Representation stack (v36). ctftshift (shift-augmented fine-tune) plus TaxaBind text delivered the largest single leap (+1.26 overall, +1.91 unseen). Seen rose to ~78.95% and has remained essentially flat through later prototypes — evidence that subsequent work is almost purely an unseen-b campaign.

External prototypes (v37–v41). Dense iNaturalist mean prototypes transferred (~0.12 overall-pt per proxy-pt). Maxpool added only +0.03 real despite large holdout deltas (~0.02 transfer). BioCLIP-2 mean prototypes stacked orthogonally (+0.08 real; ~0.08 transfer). TaxaBind image prototypes (v39) are the inverse case: positive proxy, negative real — we treat that proxy family as unreliable for EV.

Dual BioCLIP-2 (v43). Combining a frozen BioCLIP-2 prototype leg (weight 0.5) with a LoRA-v2 taxon-text leg (weight 1.0) at maxpool weight 4 produced our best measured result: 50.7556% overall, +0.186 vs v41, with unusually healthy dual-leg transfer (~0.36 overall-pt/proxy-pt). A w3 dual bracket underperformed w4 on proxy and is held.

5. Proxy discipline and failed levers

Because Codabench quota is scarce, we maintain holdout proxies and calibrate them with real anchors. Figure 5 summarizes measured transfer factors. Using the wrong factor (e.g., applying v37's 0.12 to maxpool or TaxaBind-image proxies) systematically overstates expected value and wastes slots.

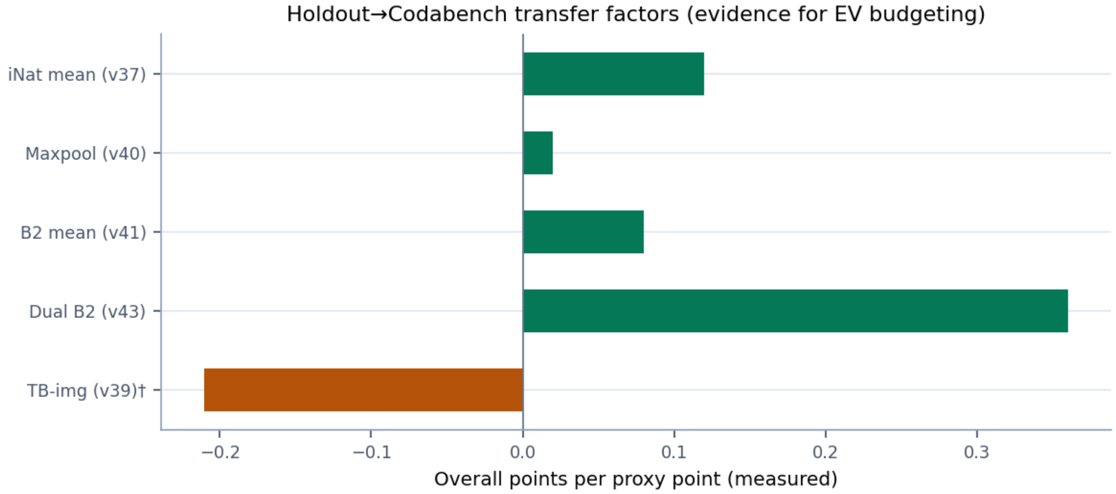

Figure 5. Measured holdout→Codabench transfer by lever family.

Lever	Holdout signal	Real outcome	Status
Recover-on-eject (v34)	+3.51 overall	–0.07 overall	DEAD
TaxaBind iNat img (v39)	+0.52 stack proxy	–0.11 overall	DEAD
SigLIP2 / DINO proto stack	< +0.3 / ~0	Not submitted	DEAD proxy
Coverage gap-fill (v38)	proxy-negative	Not submitted	DEAD
VLM top-K rerank	oracle misleading	–16 pt on n=300	DEAD
Trait / Wikipedia text	■ taxon prompts	Not load-bearing	DEAD
B2 maxpool / denser445	≤0 / worse denser	NO_ZIP	DEAD
B2 LoRA v4/v5 / hard-neg / CE	db below v1	NO_ZIP	DEAD
BioTrove-CLIP proto	–0.86 vs v41+B2	NO_ZIP	DEAD
Gated BioCLIP-2.5 B/L, ToL	401 without token	Blocked	BLOCKED

Table 3. Negative or blocked results (do not retry without new evidence).

6. Scoring decomposition and remaining headroom

For v43, population-weighted contributions are approximately 44.49 pt from seen and 6.27 pt from unseen, summing to 50.76%. At flat seen (78.952%), reaching 53% overall requires unseen accuracy of 19.497% — a +5.14 pt unseen lift. Perfect-gate oracle estimates with the current stack remain near ~53.6%, implying gate efficiency and unseen representation—not closed-set accuracy—are the binding constraints.

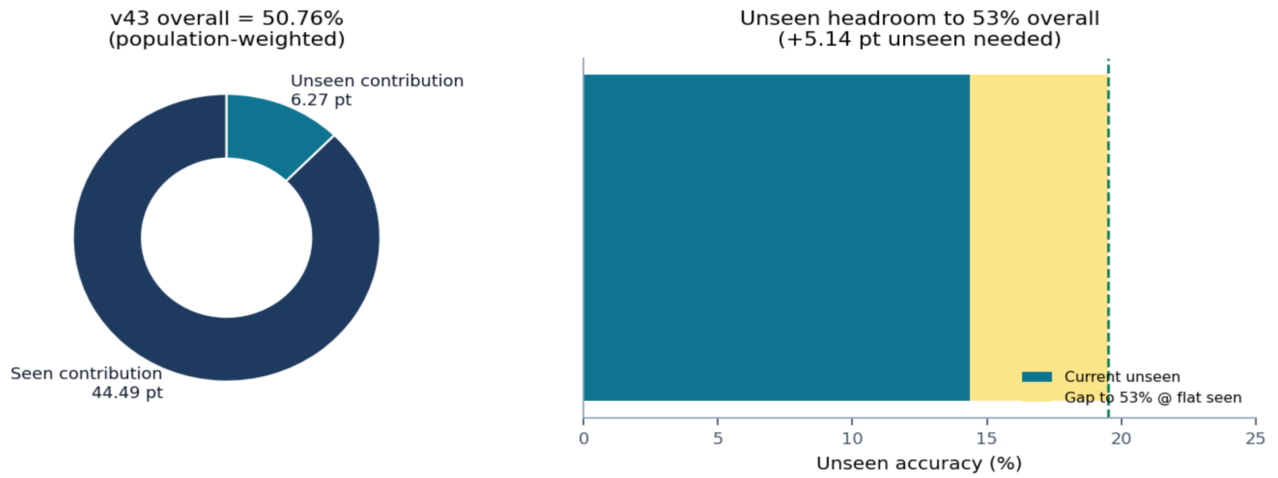

Figure 6. Score composition and unseen gap to the 53% target.

Benchmark	Overall %	Seen %	Unseen %	Δ overall vs us
Our best (v43)	50.756	78.952	14.356	—
Visible #1	50.56	78.05	15.08	–0.196 (we ahead overall)
Target bar	53.00	78.952 (flat)	19.497 (needed)	+2.244 needed
Early baseline (γ30)	45.19	—	—	–5.566 from start

Table 4. Competitive position (measured Codabench / defined targets).

7. Compliance and disclosure

We operate under the official Codabench rules. Folder-oracle routing is never used. Fish-specific third-party recognizers (e.g., CaesiumSG/sg-fish-bioclip-v2, ReefNet/finetuned-bioclip) are excluded. External morphological/taxonomy text and iNaturalist photographs used as prototypes are disclosed for the transparency statement. When model eligibility is unclear (gated general-biology encoders), we stop and ask organizers (cv4e.workshop@gmail.com) before use. If transduction is questioned, the strict single-pipeline builder `builders/build_v31_strict_singlepipeline.py` (47.69%) is the compliance reference; production best remains the dual BioCLIP-2 builder.

Rule	Our practice
No folder-oracle	One gate from image/text signals; full 17,393 argmax
No fish-only 3rd-party models	BioCLIP / general VLM family only
No test-label fishing	No manual labelling of eval images
Disclose external data	iNat photos + taxonomy text logged
Quota discipline	Submit only +EV after calibrated transfer

Table 5. Compliance checklist.

8. Rebuild instructions

Environment: `conda activate onet`. Best submission rebuild:

```
python builders/build_v43_bioclip2_dual.py
→ submissions/submission_v43_b2dual_w4_wf0.5_wl1_sink72.zip
```

Prior anchors: `build_v41_bioclip2_proto.py` (50.57%), `build_v40_inat_maxpool.py` (50.49%), `build_v36_ctftshift_taxabind.py` (49.04%), `build_v33_sinkhorn.py` (47.78%).

9. Conclusions

We advanced from ~45% to **50.76%** overall under a strict single-pipeline regime, with all late-stage progress coming from unseen-route representation (prototypes and dual BioCLIP-2) rather than seen closed-set accuracy.

Visible #1 is behind us on overall but ahead on unseen (~15.08% vs 14.36%); closing the internal 53% bar still requires approximately +5.14 pt unseen at flat seen. Local stack levers are largely exhausted; remaining plausible upside is stronger eligible general-biology encoders (pending authentication and organizer confirmation), not further low-signal weight brackets.

— End of report —

Source of truth for live scores: repository HANDOFF.md · Rules: COMPETITION_RULES.md / Codabench 16815